

18

Lect 5.1 slide 21

- divide by lowest $\wedge$ for non-beneficial
- for beneficial, divide by max
- multiply by weight

1(a)

	Price	Library size	Resolution limit	Performance
Lib 1	$\frac{2000}{2500} \times 0.2$	$\frac{16}{32} \times 0.2$	$\frac{12}{16} \times 0.4$	$\frac{5}{5} \times 0.2$
Lib 2	$\frac{2000}{2000} \times 0.2$	$\frac{16}{32} \times 0.2$	$\frac{8}{16} \times 0.4$	$\frac{3}{5} \times 0.2$
Lib 3	$\frac{2000}{3000} \times 0.2$	$\frac{32}{32} \times 0.2$	$\frac{16}{16} \times 0.4$	$\frac{4}{5} \times 0.2$
Lib 4	$\frac{2000}{2750} \times 0.2$	$\frac{32}{32} \times 0.2$	$\frac{8}{16} \times 0.4$	$\frac{4}{5} \times 0.2$

Price $\rightarrow$ non ben

Reso Lim $\rightarrow$ 40%.

Others $\rightarrow \frac{100-40}{3} = 20\%$.

	Price	Library size	Resolution Limit	Performance
Lib 1	0.16	0.1	0.3	0.2
Lib 2	0.2	0.1	0.2	0.12
Lib 3	0.1334	0.2	0.4	0.16
Lib 4	0.1454	0.2	0.2	0.16

Lib 1 $\rightarrow$ 0.76

Lib 2 $\rightarrow$ 0.62

Lib 3 $\rightarrow$ 0.8934

Lib 4 $\rightarrow$ 0.7054

(Ans)

Highest

1(b). [Lect 5.1, slide 77]

Lead

— amount of time whereby a successor activity can be advanced with respect to a predecessor activity

— e.g.

duration of A — 3 days

duration of B — 2 days

B has a fs relationship with A
• with a lead of 1 day.

A $\xrightarrow{\text{fs-1d}}$ B

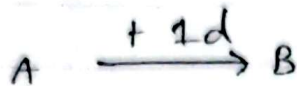
— B starts 1 day before scheduled finish of A

— So total time taken is 4 days instead of 5 days.

Lag

↳ amount of time whereby a successor activity is required to be delayed with respect to a predecessor activity.

e.g.



- B starts 1 day after the scheduled finish of A

- so, total time = 5 days instead of 4 days

1(c) [Lecture 7, slide 24]

Factors to be considered:

- ① project manager of project management team should effectively negotiate & influence others to acquire required team.
- ② failure to acquire HR for the project may affect
 - project schedule
 - budgets customer satisfaction
 - quality
 - risk etc.
- ③ alternative resources might not have required competencies.

2(a) 19 2(b)

2(b) [Lecture 7, slide 35]
expert

2(c) [Lec 5.2, slide 42]

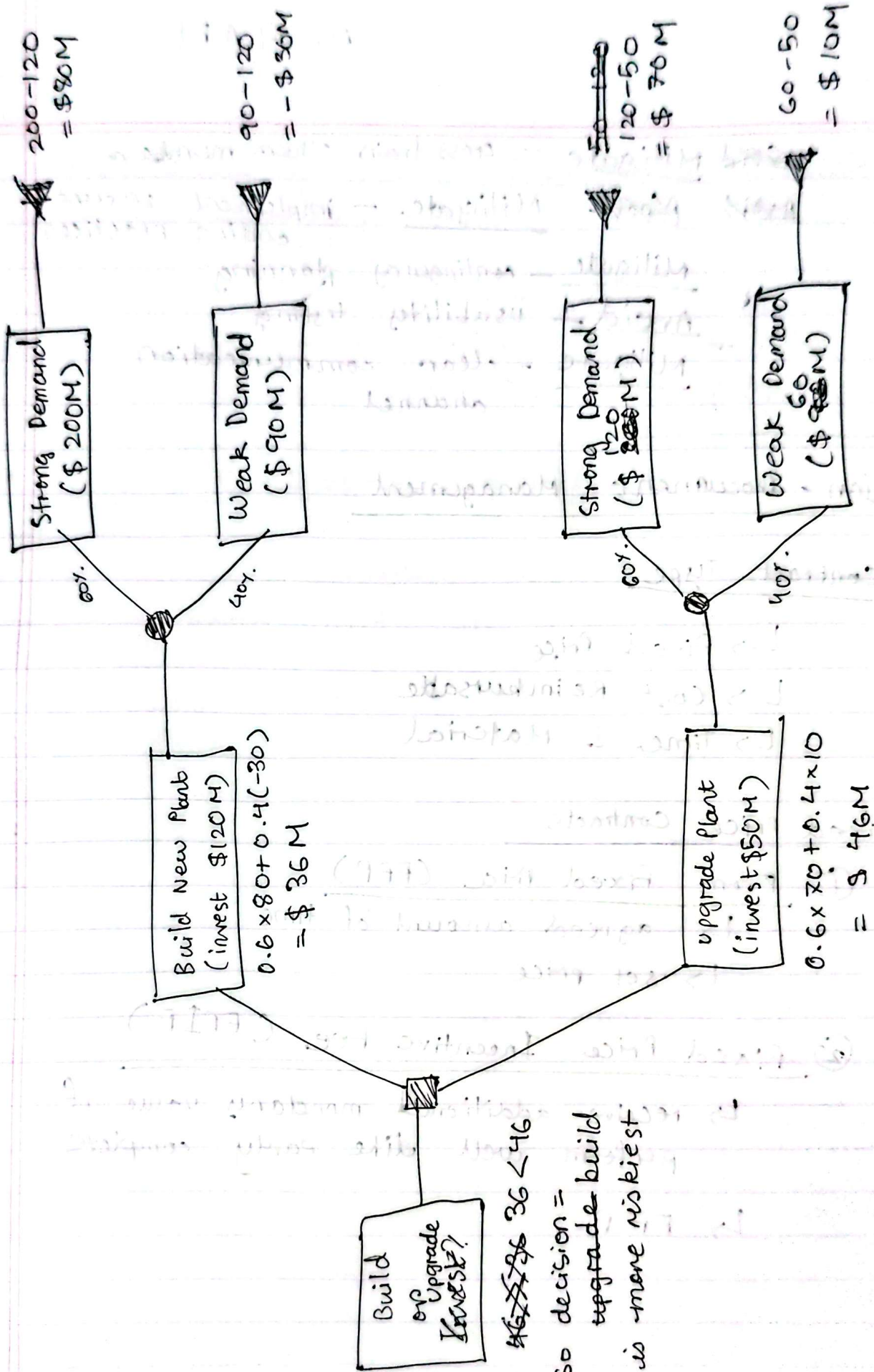
$$\text{CoC} = 50,000$$

$$\text{CoNC} = 5000 \times 50 = 25,000$$

19 - 2(a)

3(a) [Lec 6, slide 33]

~~Build or Upgrade?~~



3(b)

~~Avoid~~ Mitigate — cross train other members

~~Avoid~~ ~~Avoid~~ Mitigate — implement secure coding practices

Mitigate — contingency planning

Avoid — usability testing

Mitigate — clear communication channel

See 6,
slide 54

Plan Procurement Management

Contract Type

↳ Fixed Price

↳ Cost Reimbursable

↳ Time & Material

• Fixed Price Contracts

① Firm Fixed Price (FFP)

↳ agreed amount of time

↳ set price

② Fixed Price Incentive Fee (FPIF)

↳ receive additional monetary value if perform well like early complete

↳ FFP

$$\begin{aligned}
 PTA &= \frac{\text{Ceiling Price} - \text{Target Price}}{\text{Buyer's Ratio}} + \frac{\text{Target Total cost}}{1} \\
 &= \frac{120,000 - 112,000}{0.8} + 100,000 \\
 &= 110,000
 \end{aligned}$$

3(c)

Target cost = 2,000,000

Target price = 2,200,000

Ceiling price = 2,450,000

Ratio = 80 : 20

$$\begin{aligned}
 PTA &= \frac{2,450,000 - 2,200,000}{0.8} + 2,000,000 \\
 &= \$2,312,500
 \end{aligned}$$

4(a) [Lec 5-2, slide 67]

• Communication Methods

- ① Interactive — meeting, phone call, video conferencing
- ② Push — letter, email, memo, report
- ③ Pull — internet site, knowledge repo

4(a) project meeting $\rightarrow$ interactive
message (voicemail) $\rightarrow$ push
call $\rightarrow$ interactive
web space $\rightarrow$ pull

4(b) 19 en 4(a) FSNPA
Rec 7, slide 31

4(b) Performing stage

4(c) Rec 6, slide 54
E.g. target cost = ~~120~~ 100k
target fee = 20k
ceiling price = 130k
Share ratio = 50:50
Actual cost = ~~120k~~ 120k

target price = $100 + 20 = 120k$

cost variance = $100 - 120 = -20k$

seller's share = $0.5 \times (-20) = -10k$

fee = $20 + (-10) = 10k$

price = $120 + 10 = 130k$

So seller buyer pays 130k to seller which
is $> 100k$ & seller gets 10k fee instead
of 20k

So, fee does not ^{necessarily} become 0 at PTA.
but profit becomes ~~0~~ 0

4(d) [Lec 7, slide -4]

~~Smooth / accommodate~~ withdraw / Avoid

- she did not voice her concern &
- later expressed privately that her concerns were still valid

5(a) Lec 9, slide 15

5(b) [Lec 1, slide 12 +]

- Project

A project is a temporary endeavor, undertaken to create a unique product, service, or result.

- Program

A program is a group of projects, subsidiary programs and program activities that are managed in a coordinated manner to obtain benefits not available from managing them individually.

- Portfolio

A collection of projects, programs, subsidiary portfolios and operations managed as a group to achieve strategic objectives.

Difference

- program & project managements focus on doing programs & projects the "right" way
- portfolio management focuses on doing the "right" programs & projects

5(c) [Lect - I, slide - 35]

- ① Invisibility — with swe, progress may not be immediately visible
- ② Complexity — swe products contain more complexity
- ③ Conformity — swe developers have to conform to requirements of human clients that keep changing
- ④ Flexibility — swe can be changed.

5(d) 19 or 4(b)

[Rec 5.1, slide 76]

6(a)

deepseek

Limitations

- No Context
- No value tracking
 - Hides scope changes
- Can be Manipulated
- Pressure to conform

6(b) & (c) 19 er